

A Geometric Characterisation of Prime Numbers Using Regular Polygons and Digons

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Expository note

Abstract

Fix a unit of length. We represent a positive integer N by a regular closed route with $k \geq 2$ equal steps of integer length $s \geq 1$, so that $N = ks$. Ordinary regular convex polygons are supplemented by the degenerate digon: a segment traversed out and back. For $N \geq 2$, primality is equivalent to every admissible route having unit step length. The number of representation types is $R(N) = \tau(N) - 1$, where τ counts positive divisors. Thus 1, primes and composites have respectively zero, one and multiple types. This is an expository reformulation of elementary divisibility, using at most two spatial dimensions.

1 Motivation

This construction arose from an attempt to develop a geometric language for prime numbers as a preliminary step towards investigating additive questions such as Goldbach's conjecture; no result concerning Goldbach's conjecture is claimed here. Encoding factorisation geometrically does not by itself resolve an additive question. The present purpose is to state the correspondence precisely and treat the small integers uniformly.

2 Definitions

We work throughout with *positive integers*, and fix a unit of length. A prime is an integer greater than 1 whose only positive divisors are 1 and itself. A composite is an integer greater than 1 that is not prime; 1 is neither prime nor composite.

Definition 1. An *admissible regular closed route* of total length N has k equal straight steps of length s , where $k \geq 2$ and $s \geq 1$ are integers. For $k \geq 3$ it traverses the boundary of a regular convex k -gon once. For $k = 2$ it traverses a segment of length s and retraces it to the starting point; we call this a *degenerate digon*. Its total travelled length is $N = ks$.

The digon is a convention for a closed route, not an ordinary non-degenerate Euclidean polygon. Its two traversals coincide: the geometric segment has length s , the travelled distance is $2s$, and the final displacement is zero. No integer-coordinate condition is imposed. A *representation type* is determined solely by (k, s) . Position, orientation, starting vertex and direction of traversal do not create new types; in particular, rotations, translations and reflections are not counted again. Star polygons and repeated circuits are excluded.

3 Geometric characterisation of primality

Every pair of integers $k \geq 2$, $s \geq 1$ is realisable: for $k \geq 3$, inscribe a regular k -gon in a circle of radius $s/(2 \sin(\pi/k))$; for $k = 2$, use the out-and-back route. In particular, every $N \geq 2$ has the unit-step type $(N, 1)$.

Theorem 2. *For every integer $N \geq 2$, N is prime if and only if every admissible regular closed route of total length N has unit step length. Equivalently, no such route has integer step length $s > 1$.*

Proof. If an admissible route has $s > 1$, then $N = ks$ with $k, s \geq 2$, so N is composite. Conversely, if N is composite, write $N = ab$ with $a, b \geq 2$. Choose $k = a$ and $s = b$. When $a = 2$, use the digon; when $a \geq 3$, use a regular convex a -gon of side length b . This is an admissible route with non-unit integer step length. These two implications prove the equivalence. If N is prime, $k \mid N$ and $k \geq 2$ force $k = N$, and hence $s = 1$; its unique representation type is $(N, 1)$. $\square$

The assumption $N \geq 2$ is essential: 1 admits no route, so the universal unit-step condition would hold vacuously at 1.

4 The exceptional role of the digon

If only ordinary polygons ($k \geq 3$) were allowed, the same criterion would work for $N \geq 5$. Indeed, a composite $N \geq 5$ has a factorisation $N = ab$ with $a \geq b \geq 2$ and $a \geq 3$: otherwise $a = b = 2$ and $N = 4$. At 4, however, the only ordinary-polygon type is the unit square $(4, 1)$; the missing non-unit witness is $(2, 2)$. Admitting the digon provides this witness and also gives the prime 2 its unit route $(2, 1)$. Thus $2 = 2 \cdot 1$ and $4 = 2 \cdot 2$ fit the same convention.

Equal turns. Each route can be followed using the same oriented exterior turn $360^\circ/k$ after each edge, including at the closing vertex. For $k = 2$ the 180° turn reverses direction along the segment; for $k = 3$ the 120° turn produces an equilateral triangle. This unifies the construction geometrically without enlarging the admissible family to other turning angles or star polygons.

5 Representation count and classification

For a positive integer N , define

$$R(N) = \#\{(k, s) \in \mathbb{Z}^2 : k \geq 2, s \geq 1, ks = N\}.$$

Let $\tau(N)$ be the number of positive divisors of N (also denoted $d(N)$ in standard references [1]).

Proposition 3. *For every positive integer N , $R(N) = \tau(N) - 1$.*

Proof. Every divisor $k \mid N$ with $k \geq 2$ gives exactly one type $(k, N/k)$, and every type arises in this way. Of the $\tau(N)$ positive divisors, only $k = 1$ is excluded. $\square$

Corollary 4. *For positive integers N ,*

$$N = 1 \iff R(N) = 0, \quad N \text{ prime} \iff R(N) = 1, \quad N \text{ composite} \iff R(N) \geq 2.$$

Proof. The integer 1 has no type. Every $N \geq 2$ has exactly one unit-step type, $(N, 1)$; Theorem 2 shows that an additional type exists exactly when N is composite. $\square$

6 Examples

Table 1 lists all types for $1 \leq N \leq 12$. The two factors have different roles, so, for example, $(3, 4)$ and $(4, 3)$ are distinct types. Figure 1 makes the distinction visible for $2 \leq N \leq 6$.

Table 1: All representation types for the first twelve positive integers.

N	Representation types (k, s)	Classification
1	none	neither prime nor composite
2	$(2, 1)$	prime
3	$(3, 1)$	prime
4	$(2, 2), (4, 1)$	composite
5	$(5, 1)$	prime
6	$(2, 3), (3, 2), (6, 1)$	composite
7	$(7, 1)$	prime
8	$(2, 4), (4, 2), (8, 1)$	composite
9	$(3, 3), (9, 1)$	composite
10	$(2, 5), (5, 2), (10, 1)$	composite
11	$(11, 1)$	prime
12	$(2, 6), (3, 4), (4, 3), (6, 2), (12, 1)$	composite

Regular routes for total lengths 2 to 6

Primes have one representation type; composites have more.

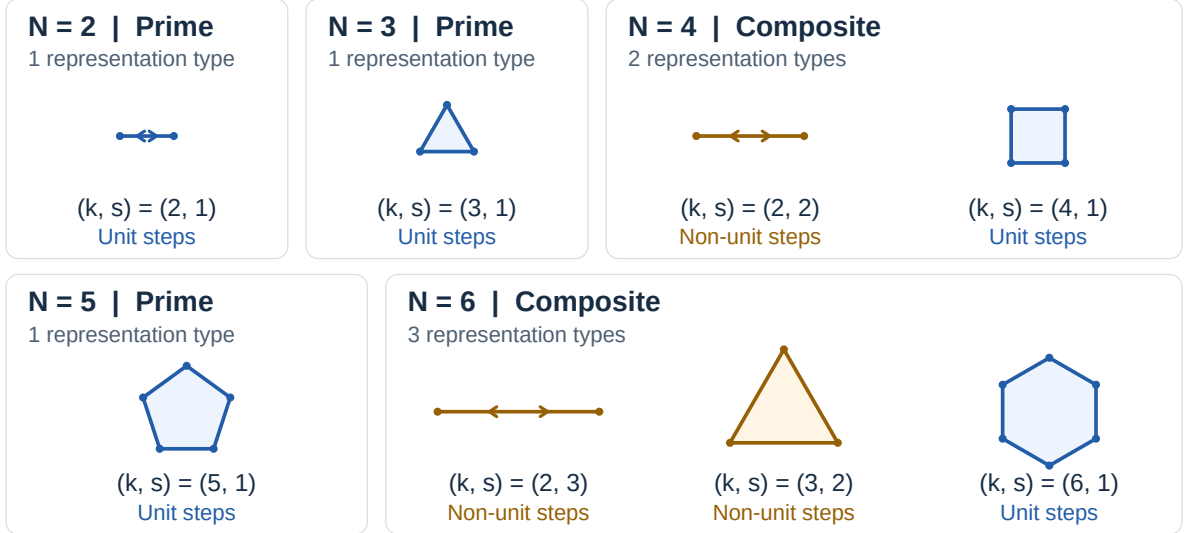


Figure 1: All types for $N = 2, 3, 4, 5, 6$, at a common length scale. Blue routes have $s = 1$; amber routes have $s > 1$, also indicated by the labels. The digon's opposing arrows lie on a single segment, which is traversed twice. Primes have one type; composites have more.

7 Dimensional scope

The digon lies in one dimension, while ordinary regular polygons lie in a plane. The construction therefore requires at most two spatial dimensions. Every planar route embeds unchanged in $\mathbb{R}^n$ for $n \geq 2$, so the result imposes no upper bound on mathematical or physical dimensions. Higher-dimensional extensions are left for separate investigation; this note establishes only the planar case.

8 Relation to existing geometric descriptions of primality

Geometric descriptions of factorisation and primality already exist. Rectangular arrays represent products by rows and columns; Way's educational treatment explicitly connects their shapes with factors and prime numbers [3]. Polygonal and star-polygon constructions also express divisibility and coprimality: Erfle describes how a fixed jump among cyclically arranged vertices relates to whether a star is traced in one circuit [2]. Those constructions provide context; star polygons are not routes admitted by our definition.

This note records an independently developed geometric reformulation of elementary divisibility. The correspondence between geometric representation and factorisation is not claimed to be a new result in number theory. The purpose is to formulate the total-length construction explicitly, remove the exception caused by ordinary polygon terminology by admitting the degenerate digon, and derive a uniform representation count. Historical priority for this exact formulation is not claimed; whether this precise presentation has appeared before has not been established. The references are contextual, not an exhaustive literature or priority investigation.

9 Conclusion

Factorisation appears as alternative regular closed routes with equal integer step lengths. Primes have one representation type, composites have multiple types, and the total number is $\tau(N) - 1$. The digon makes the primality criterion uniform from 2 upwards, while 1 is distinguished by having no representation. The result is a geometric description of the prime/composite distinction, with its arithmetic content made explicit.

Source and licence

The source, vector diagrams and finite arithmetic checks are available at the accompanying repository. The checks guard against transcription errors; the proofs above establish the general results. Copyright © 2026 Pranay Manocha. The note and artwork are licensed under Creative Commons Attribution 4.0 International.

References

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